DATA 8

Spring 2024

Lecture 34

Updating Probabilities

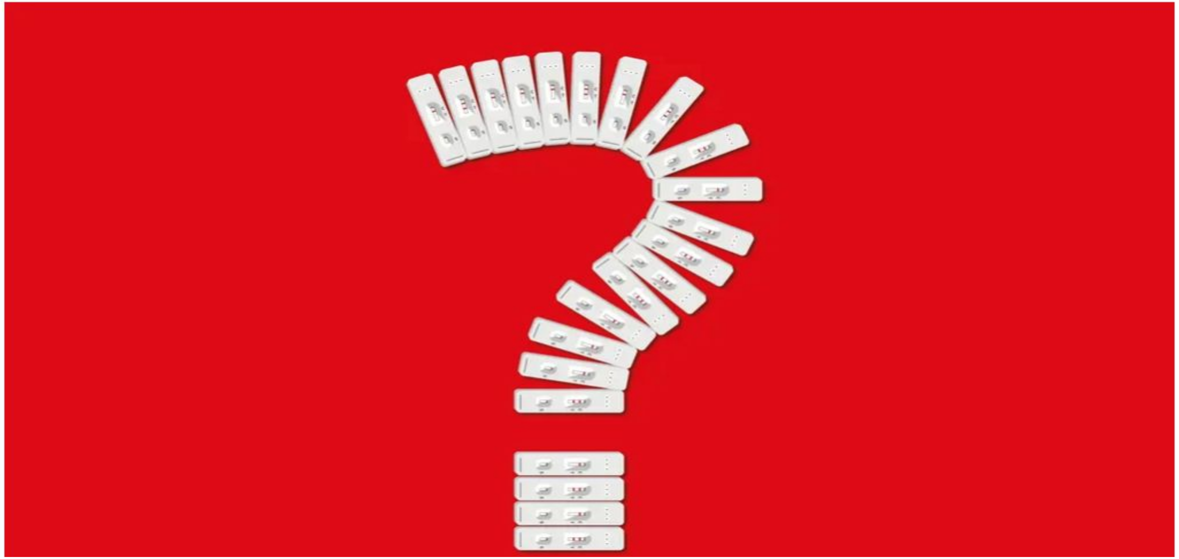
Announcements

- **Lab 9** due today at **5 pm**
- **Project 2** due today at **5 pm**
- **Project 3** coming soon

Updating Probabilities

The obscure maths theorem that governs the reliability of Covid testing

[Source: Guardian]



There's been much debate about lateral flow tests – their accuracy depends on context and the theories of a 18th-century cleric

> *Without knowing the prior probability, you don't know how likely it is that a result is false or true*

The government says – accurately – that the "false positive rate", the chance of a test returning a positive result in a person who does not have the disease, is less than one in 1,000. And that's where we came in: you might think that that means, if you've had a positive result, that there's a less than one in 1,000 chance that it's false.

A Question at a Medical School

Interpretation by Physicians of Clinical Laboratory Results (1978)

> "We asked 20 house officers, 20 fourth-year medical students and 20 attending physicians, selected in 67 consecutive hallway encounters at four Harvard Medical School teaching hospitals, the following question:
>
> "If a test to detect a disease whose prevalence is 1/1000 has a <u>false positive rate</u> of 5%, what is the chance that a person found to have a positive result actually has the disease, assuming that you know nothing about the person's symptoms or signs?"

Rate of Correct Answers

Interpretation by Physicians of Clinical Laboratory Results (1978)

> "Eleven of 60 participants, or **18%, gave the correct answer**. These participants included four of 20 fourth-year students, three of 20 residents in internal medicine and four of 20 attending physicians. The most common answer, given by 27, was that the chance that a person found to have a positive result actually has the disease was 95%.

Conditional Probability

A Simpler Setting

- Scenario:
  - Class consists of second years (60%) and third years (40%)
  - 50% of the second years have declared their major
  - 80% of the third years have declared their major
- **I pick one student at random.**
- Which is more likely: Second year or Third year?
  - Second year, because they are 60% of the class

Unconditional Probabilities

- Scenario:
  - Class consists of second years (60%) and third years (40%)
  - 50% of the second years have declared their major
  - 80% of the third years have declared their major
- **I pick one student at random.**
- P(2nd year)

  = proportion of second years **among all students**

  = 0.6
- **Unconditional probability**, given no additional information

Conditional Probabilities

- Scenario:
    - Class consists of second years (60%) and third years (40%)
    - 50% of the second years have declared their major
    - 80% of the third years have declared their major
- **I pick one student at random … They are in their second year!**
- P(declared their major given they are in their second year)

  = proportion declared **among all second years** = 0.5
- **Conditional probability**, given that the student is a second year
- Denoted P(Declared | 2nd year) [vertical bar: “given”]

Same Question, More Information

- Scenario:
  - Class consists of second years (60%) and third years (40%)
  - 50% of the second years have declared their major
  - 80% of the third years have declared their major
- **I pick one student at random...**
  That student has declared a major!

(Demo)

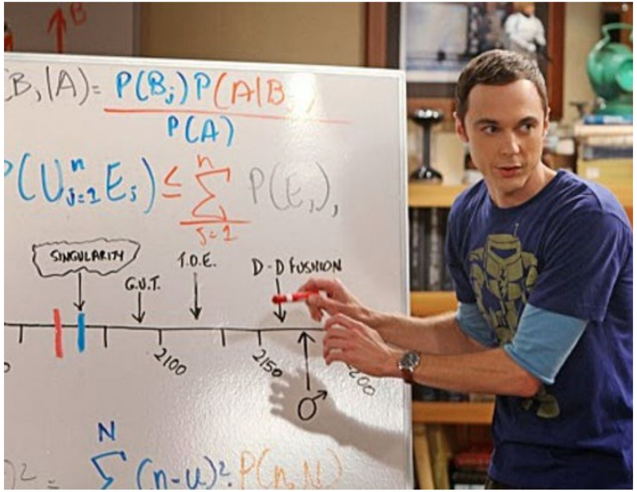
- Which is more likely: Second Year or Third Year?
- Now we need P(2nd year | Declared)
- = proportion of second years **among all those who have declared**

Bayes' Rule

Purpose of Bayes' Rule

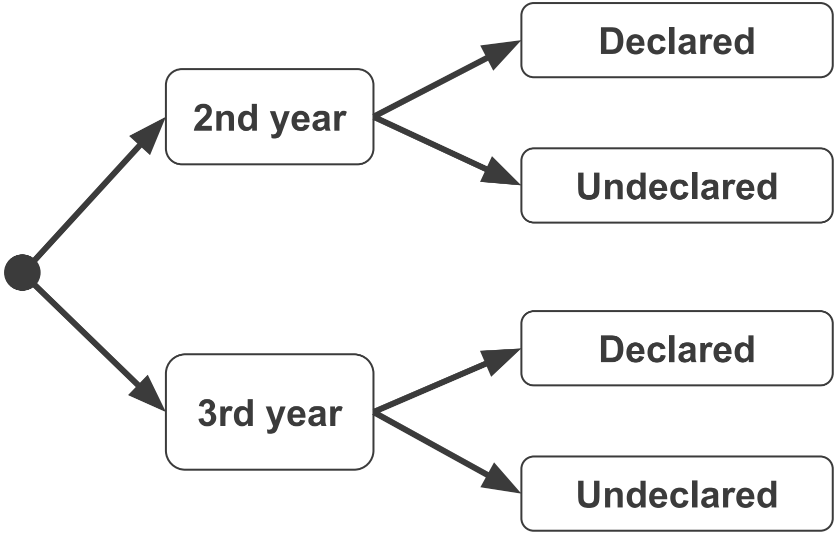
- Update your prediction based on new information

- In a multi-stage experiment,
  find the chance of an event at an earlier stage,
  given the result of a later stage



Tree Diagrams

Depicts each event and the probability of the next event **given** the current event.

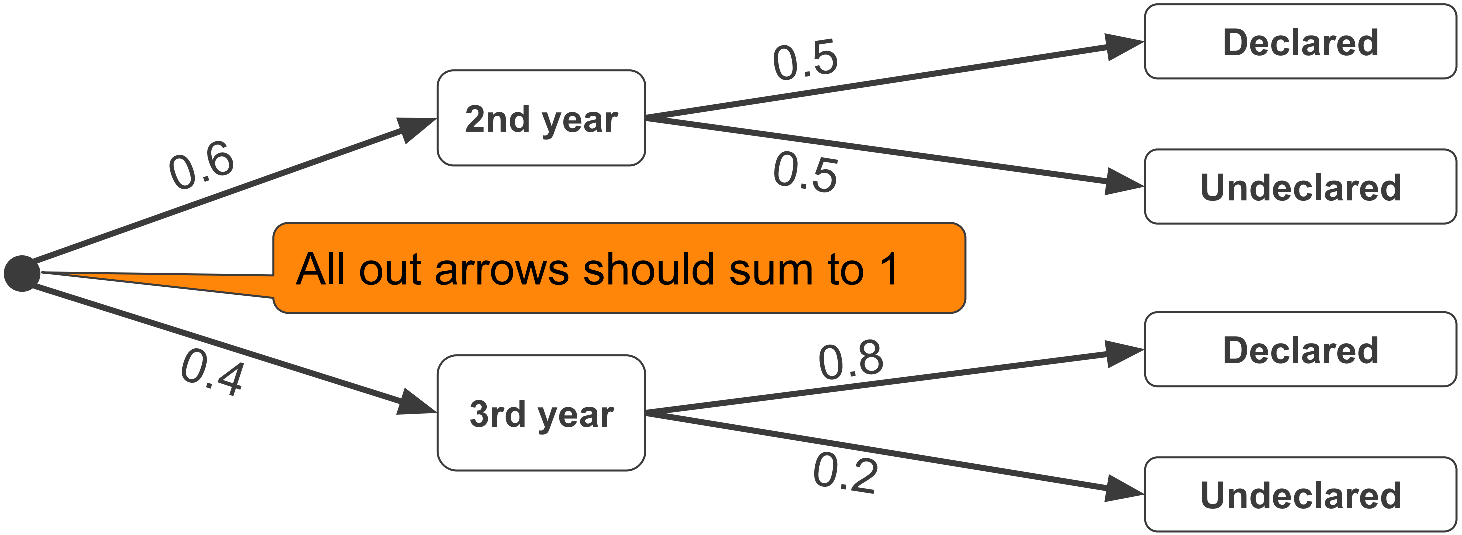


This is called a tree diagram because it sort of looks like a tree if you plant your trees sideways…

Tree Diagram

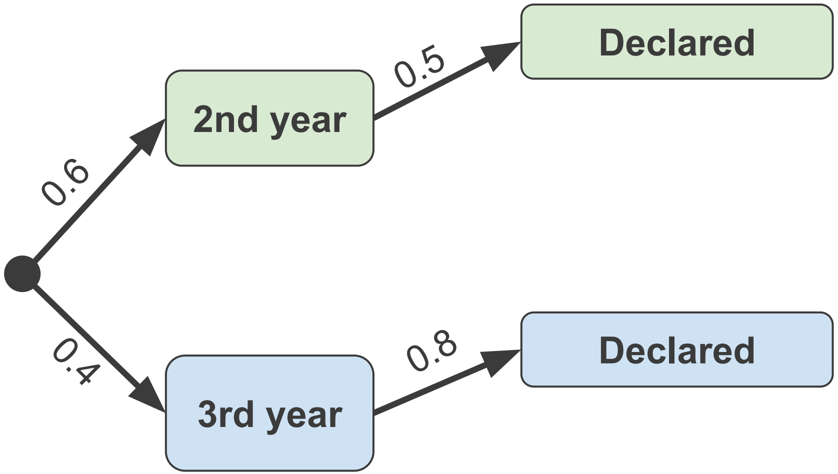
Depicts each event and the probability of the next event **given** the current event.

For a randomly selected student:



Working With the Tree

- Start by identifying the branches where the event in this case, declared) occurs.



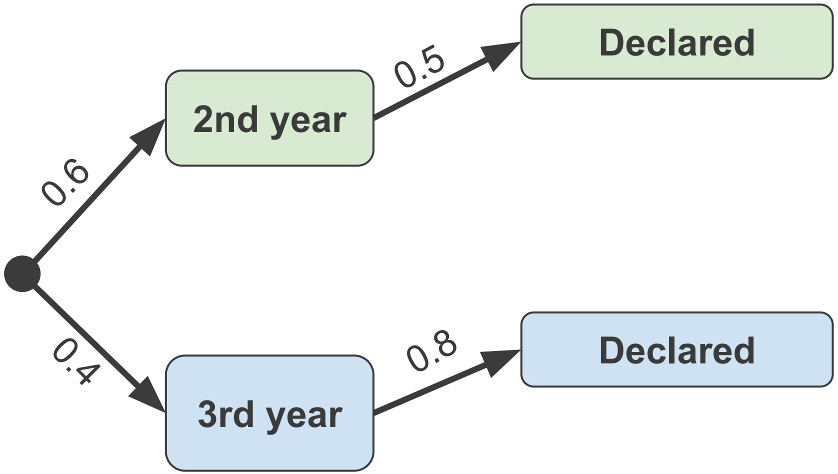
P(2nd year and Declared) = 0.6 x 0.5 = 0.3

P(3rd year and Declared) = 0.4 x 0.8 = 0.32

P(Declared) = 0.3 + 0.32 = 0.62

Bayes Calculation

For finding chances at a first stage, given the result of a second stage.



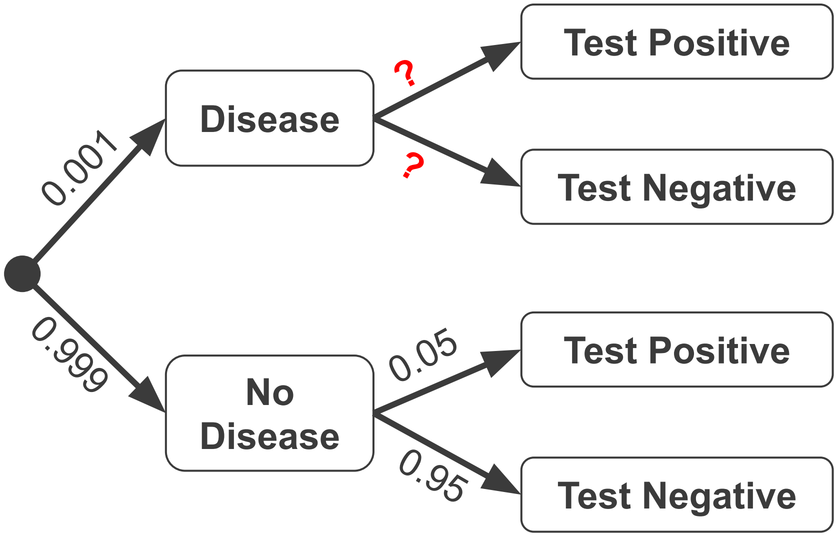
P(2nd Year | Declared)

$$= \frac{0.6 \times 0.5}{(0.6 \times 0.5) + (0.4 \times 0.8)}$$

$$= \frac{0.3}{0.62} = 0.4839\ldots$$

Doctors and Clinical Tests

- Back to the Harvard Medical School example

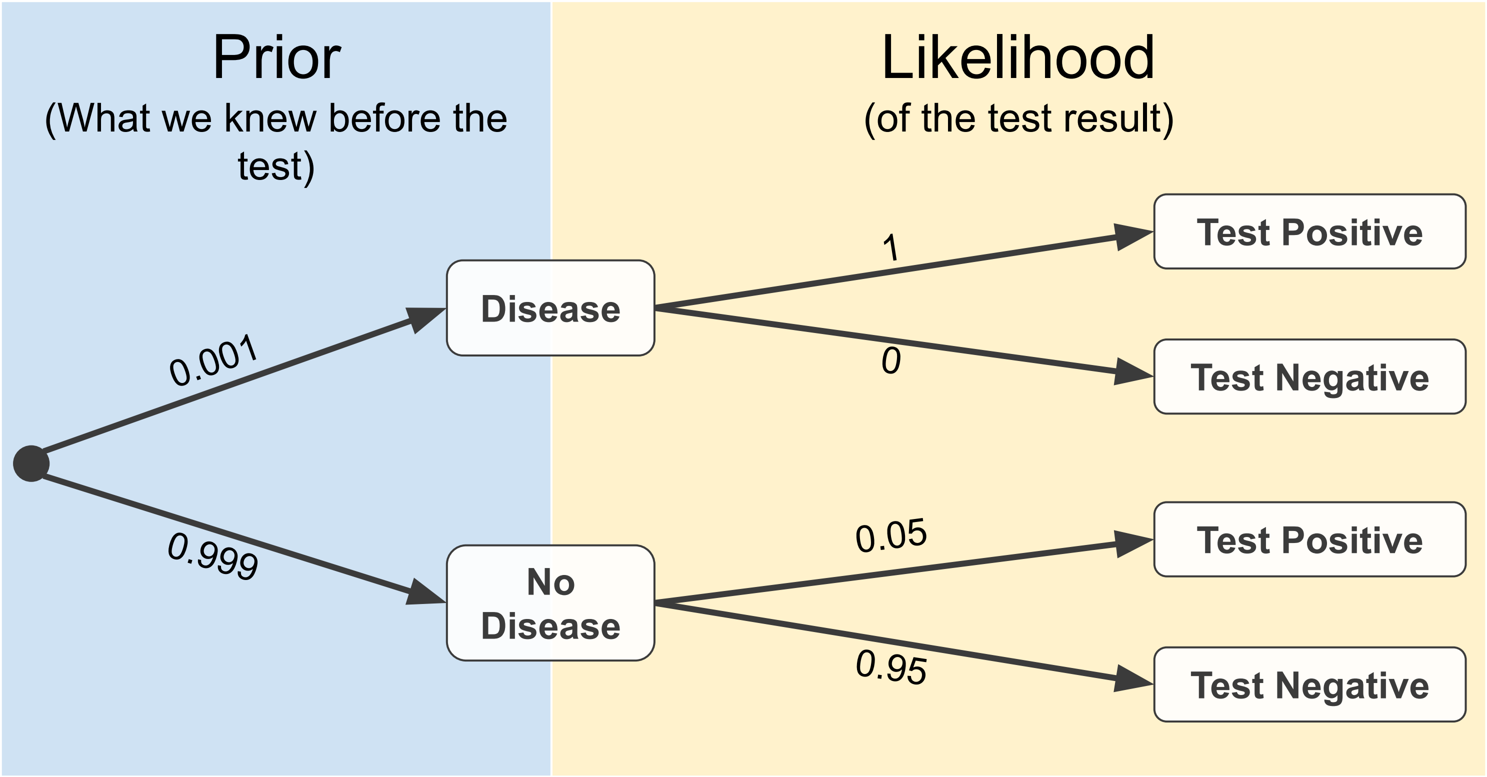


Problem did not give the *true positive* rate.

That's the chance the test says "positive" if the person has the disease.

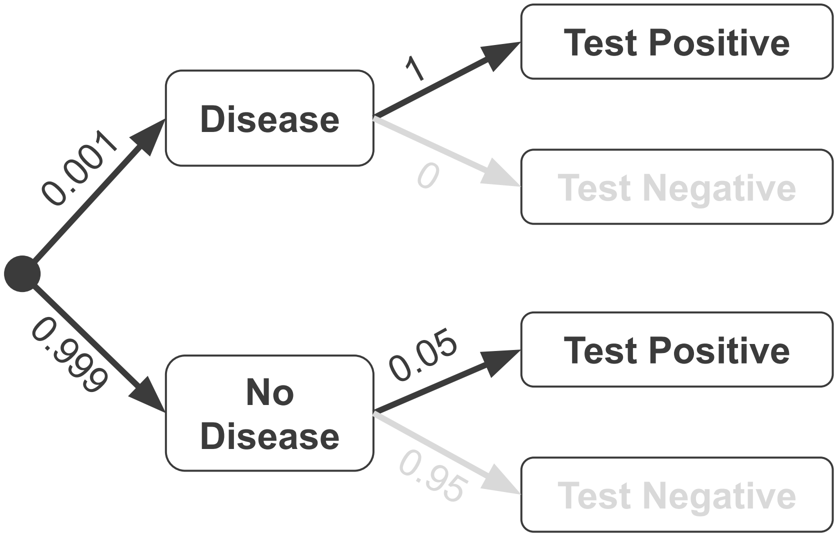
It was assumed to be 100%.

Tree Diagrams and Terminology



Posterior Probability

Updated chance of an event given the result of a later stage



P(Disease | Test Positive)

$$= \frac{0.001 \text{ x } 1}{(0.001 \text{ x } 1) + (0.999 \text{ x } 0.05)}$$

$$= \frac{0.001}{0.001 + 0.04995} = 0.01963$$

(Demo)

A Closer Look at the Answer

Assume a patient is picked at random.

- Prior probability of disease
  - P(Disease) = 0.001 = one-tenth of 1%
- Posterior probability of disease given positive test
  - P(Disease | Test Positive) = 0.0196... ≅ 2%
- Bigger than the prior, but still pretty small
- Should we approve such a test?
  - The test has **low error rates** compared to most tests
- How can this be?

Assumptions Matter

- **“Assume a patient is picked at random.”**
  - But usually, people aren’t picked at random for medical tests
  - So our intuition about randomly picked patients may not be great
- For a randomly picked patient, the result does make sense, because the disease is very rare.
- What if the doctor believes there is a 10% chance the patient has the disease?

(Demo)

Subjective Probabilities

Opinions

The probability of an outcome can be defined in two ways:

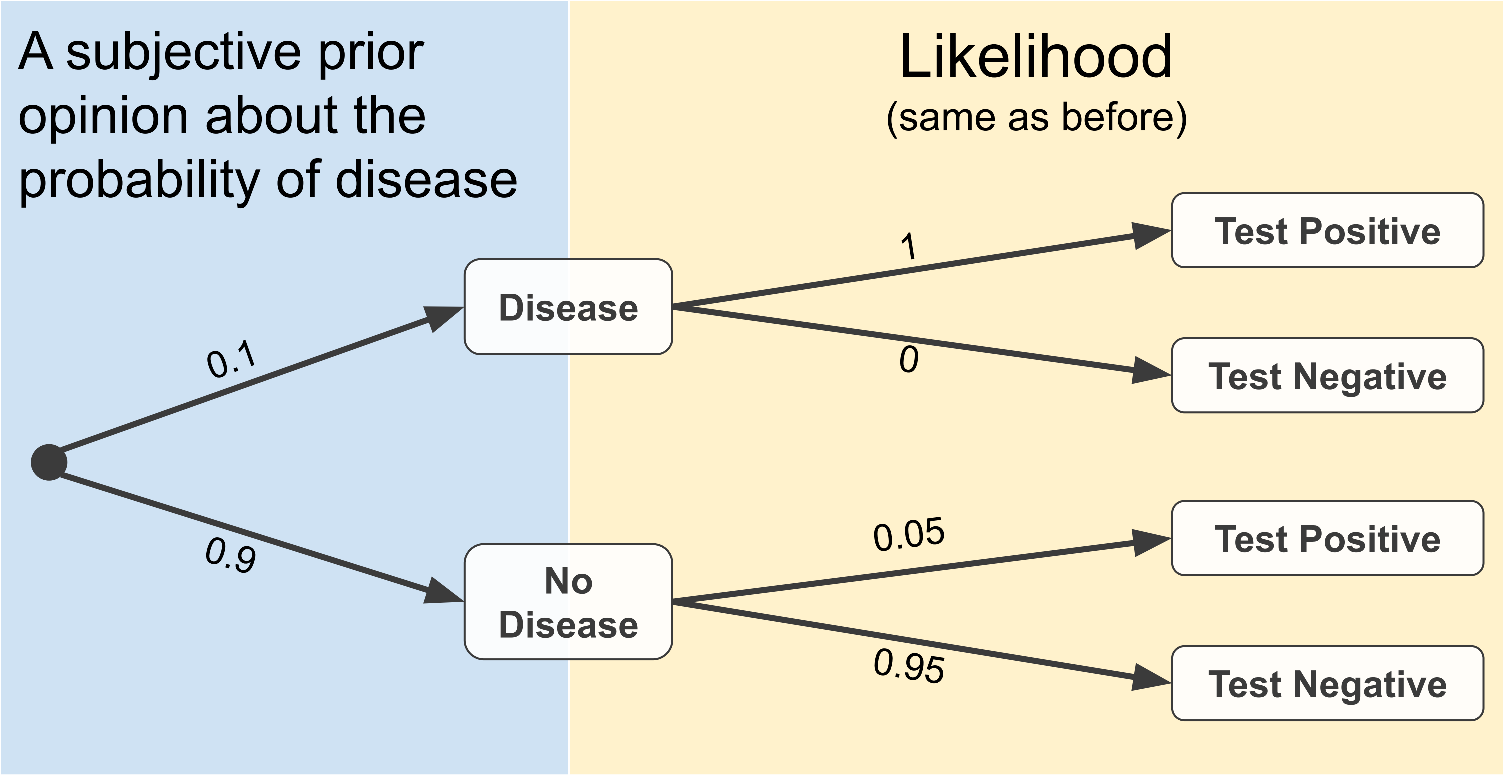
[Frequentist] The **frequency** with which it will occur in repeated trials

[Bayesian] Someone's **subjective degree** of belief that it will occur

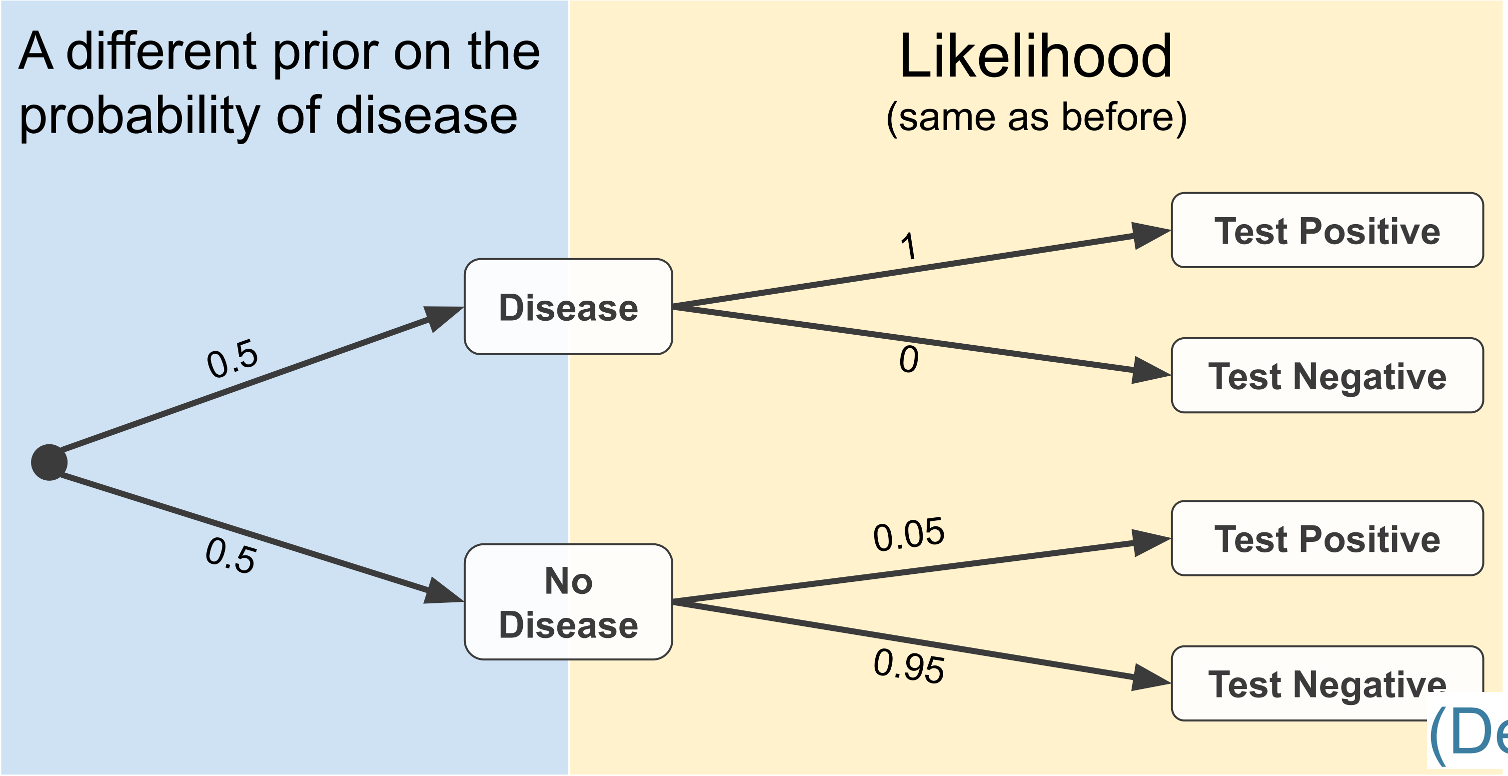
Why use subjective probabilities?

- To quantify your degree of uncertainty about an outcome, even when there is no physical randomization
  - chance of the "Big One" in the next 30 years

One Subjective Opinion



A Different Subjective Opinion



(Demo)

Discussion Question

A car heading from Berkeley to San Francisco is pulled over on the freeway **for speeding**. Which type of car is it more likely to be:

- a **Tesla** which is **relatively common** in the area
- or a **Lamborghini** which is **a rare** car that is **known for speeding**

You don’t have enough information to **calculate the answer directly.**

What would you guess, and why? Make some reasonable assumptions (data scientists often have to do this) and explain your thought process.